

Student Dropout Risk Prediction: Fair & Explainable Early Intervention

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This paper presents a fairness-aware machine learning system that identifies university students at risk of dropping out after their first semester, in order to trigger early academic and financial support interventions. Using the UCI Student Dropout dataset [1], we train an interpretable Logistic Regression baseline and a higher-performing XGBoost classifier, both constrained to use only information available at the end of semester one. We conduct a fairness audit across five sensitive attributes — gender, scholarship status, debtor status, displaced status, and age group — finding that scholarship holders face the largest false negative rate disparity (FNR gap 0.267 for XGBoost; 0.336 for Logistic Regression), meaning at-risk scholarship students are systematically under-flagged. We apply three mitigation strategies to the Logistic Regression baseline and show that Reweighting nearly eliminates the scholarship gap (FNR difference 0.008 on the LR model) with modest accuracy cost. SHAP and LIME explanations confirm that first-semester academic performance is the dominant driver of predictions. The paper concludes with a structured model card and an honest assessment of limitations and deployment considerations.

INTRODUCTION & ORGANISATIONAL DECISION

Student dropout is one of the most consequential outcomes in higher education — for institutions, funding bodies, and above all for the students themselves [2]. This project operationalises early identification of at-risk students as a machine learning pipeline.

The **organisational decision** we address is: *which students should be prioritised for proactive support after semester 1, based on predicted dropout risk?* The model triggers early intervention — academic tutoring, financial counselling, and mental health support — allocated by university advisors to flagged students.

The asymmetry of errors is critical. A **false negative** — an at-risk student not flagged — carries the highest cost, because that student may drop out precisely because no support was offered. A false positive creates a lesser cost: unnecessary stigma and wasted counselling resources. Our entire modelling and fairness strategy is structured around this asymmetry: we optimise for recall and audit fairness using False Negative Rate (FNR) as the primary metric.

Stakeholder Map

Five stakeholder groups shape this decision. University administration *decides* — it authorises the programme and sets intervention thresholds. Students are *affected*, receiving or being denied support based on model output. Academic advisors *implement* recommendations, delivering tutoring and financial support. The Data Protection Officer *audits* GDPR compliance, and the Ministry of Education evaluates dropout KPIs and may condition funding on retention outcomes.

WHAT CHANGED AFTER THE DESIGN REVIEW

This section documents changes made following the collective design review session. It is a required element of the final deliverable, as feedback from the design review is binding.

Dropping the Enrolled Class

Original framing: We initially proposed merging **Enrolled** with **Graduate** into a single Non-dropout class, retaining all 4,424 rows.

Feedback received: The professor challenged this decision, noting that the **Enrolled** class is fundamentally ambiguous. Some enrolled students are on track; others are quietly at risk. Critically, certain demographic groups — notably evening and working students — are structurally over-represented in the **Enrolled** class for reasons unrelated to academic performance. Including them as Non-dropout would teach the model a systematically incorrect signal for those groups.

What we changed: We dropped all 794 **Enrolled** rows, reducing the dataset from 4,424 to 3,630 students. This yields a clean binary problem: Graduate (0) vs. Dropout (1), with a 39% dropout rate.

Feature Timing — End of Semester 1

Original framing: Our initial proposal described the decision point as “at the start of the academic year,” implying enrollment-time features only.

Feedback received: Using only enrollment-time features would flag students on background characteristics alone —

before any academic signal exists — risking stigmatisation based on socioeconomic background.

What we changed: We updated the decision point to end of semester 1. Our feature set now includes all enrollment-time variables plus the six semester 1 academic performance columns, while explicitly excluding all semester 2 columns. End of semester 1 is the earliest point at which the university has both demographic context and real academic signal.

DATASET & EXPLORATORY DATA ANALYSIS

Dataset

We use the UCI ML Repository dataset “Predict Students’ Dropout and Academic Success” [1], collected from a real Portuguese higher education institution. After dropping **Enrolled** rows and semester 2 columns, the working dataset contains 3,630 students and 28 features. The dataset was chosen because it originates from a real administrative record (not a toy dataset), contains rich demographic variables enabling multi-dimensional fairness analysis, and has not been extensively studied from a fairness perspective.

Sensitive Attributes & EDA Findings

We identified five sensitive attributes based on group size, conceptual relevance, and harm hypotheses from our framing document. Figure 1 shows the raw dropout rates for each group before any model is trained — disparities that are *structural*, existing independent of any modelling choice.

Figure 2 shows the absolute number of students in each sensitive attribute group, providing context for interpreting per-group error rates.

Gender shows male students dropping out at 56% vs. 30% for female students — a substantial raw gap that persists into the model’s fairness results. This does not simply mean male students need more support; it means any model trained on this data will learn gender as a risk signal, and we must audit whether that signal produces equitable false negative rates across groups.

Scholarship status is the strongest *protective* socioeconomic signal in the dataset: non-scholarship students drop out at 48% vs. 14% for scholarship holders. The protective effect of scholarships is real and the model correctly learns it — but as the fairness audit reveals, this same learning causes at-risk scholarship holders to be systematically under-flagged.

Debtors drop out at 76% — the largest raw rate gap by far. Financial hardship is a genuine predictor of dropout, and we include Debtor as both a model feature (condi-

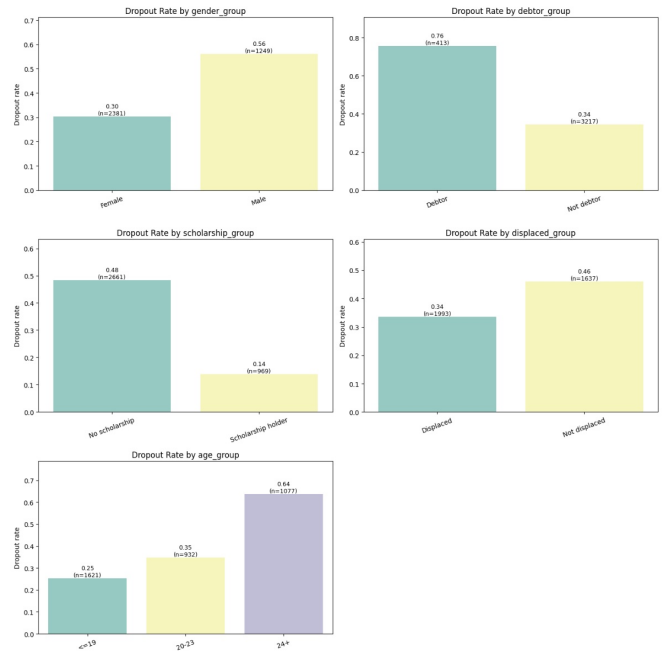


FIG. 1. Raw dropout rates by sensitive attribute group before any modelling. Debtors (76%) and students aged 24+ (64%) face the highest raw risk, followed by male students (56%). Scholarship holders (14%) show the lowest rate. These structural disparities motivate the fairness audit.

tionally, if it adds signal beyond scholarship status) and a sensitive attribute in the fairness audit.

Displaced students show a *lower* dropout rate (34%) than non-displaced students (46%). This contradicts the intuitive assumption that geographic displacement increases risk, and suggests that displaced students in this institution may self-select for stronger academic motivation, or benefit from dedicated support programmes. We retain Displaced as a sensitive attribute regardless, since the audit is about equitable false negative rates, not simply about which group has higher raw risk.

Nationality was excluded due to sparsity. Age was included as a binned variable (≤ 19 , 20–23, 24+), with students aged 24+ showing the highest dropout rate (64%).

MODEL TRAINING

Logistic Regression Baseline

We begin with a statistically grounded Logistic Regression baseline, serving both as an interpretable model and as the base estimator for fairness mitigation.

Feature selection: We first apply iterative VIF filtering (threshold 5.0), protecting sensitive attribute columns from automatic removal. This reduces the feature set from 28 to 18 candidates. We then fit a statsmodels Logit and retain only the 8 features significant at

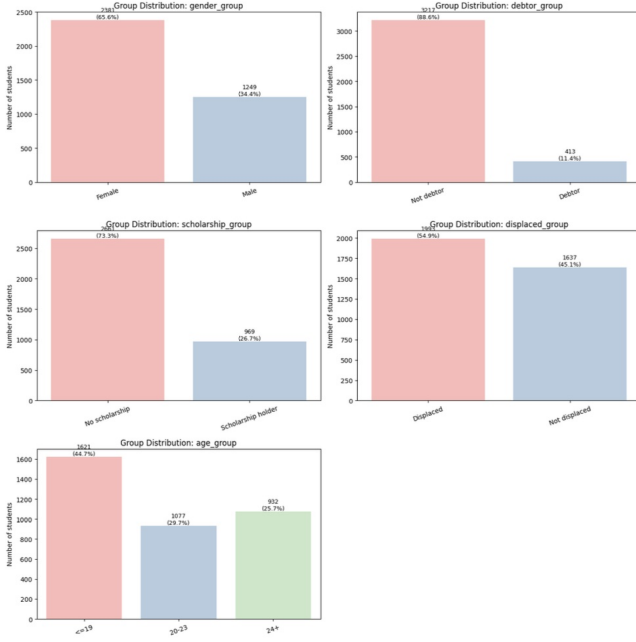


FIG. 2. Group sizes across all five sensitive attributes. Non-scholarship students (2,987) and non-debtors (3,182) represent large majority groups, while scholarship holders (969) and debtors (448) are smaller minorities. Group size disparities directly affect the reliability of per-group fairness estimates.

$p < 0.05$. The strongest effects are Debtor (odds ratio 7.89 — strongest risk factor), Scholarship holder (OR 0.29 — strongly protective), Gender (OR 1.94 — male at higher risk), and Curricular units 1st sem (grade) (OR 0.78 — protective). Logistic Regression uses `class_weight='balanced'` to account for class imbalance.

Performance: Mean weighted CV F1 = 0.798 ± 0.019 (10-fold), confirming stable generalisation.

Note on VIF filtering: Curricular units 1st sem (approved) was removed from the LR model due to high multicollinearity (VIF 19.8). SHAP subsequently identifies this as the single most important global predictor for XGBoost, illustrating that the VIF-based preprocessing required for stable LR coefficients comes at a cost: the most powerful predictive signal available is excluded from the interpretable baseline.

XGBoost

XGBoost is trained as the higher-performing non-linear comparator, using the full 28-feature set (no VIF filtering), shallow trees (max depth 4), early stopping on a held-out 20% validation split (best iteration: 173), subsampling (0.8), and L2 regularisation ($\lambda=2$). Conservative hyperparameters and early stopping are critical: with 500 candidate trees, without regularisation the model would

easily overfit the 2,541-student training set. The combination of subsampling, column sampling, and early stopping ensures the model generalises to the test set rather than memorising training patterns.

Model Comparison & Threshold Selection

Model	Recall	Prec.	F1	AUC
XGBoost	0.836	0.901	0.867	0.942
LR	0.728	0.743	0.735	0.875

TABLE I. Model performance at default threshold (0.5), test set $n = 1089$.

Figure 3 shows the XGBoost confusion matrix, prediction distributions, and ROC curve. Only 70 true dropouts are missed, compared to 116 for Logistic Regression.

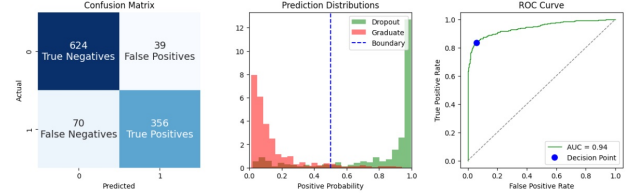


FIG. 3. XGBoost evaluation at default threshold: confusion matrix (left), prediction distributions (centre), and ROC curve with AUC = 0.94 (right).

A default threshold of 0.5 is suboptimal when false negatives carry higher cost. Using F2 score ($\beta=2$, weighting recall more heavily), we select operational thresholds: XGBoost at 0.35 (recall 0.873, precision 0.844, F2 0.867) and LR at 0.25 (recall 0.937, precision 0.545, F2 0.819).

FAIRNESS AUDIT

We audit both models using `fairlearn.metrics.MetricFrame`, which computes disaggregated metrics across sensitive attribute groups — functionally equivalent to the Group and Bias reports produced by Aequitas. All metrics are computed at the F2-selected operational thresholds. We report disaggregated results for all five sensitive attributes — gender, scholarship status, debtor status, displaced status, and age group — and focus on **False Negative Rate difference** as the primary fairness metric throughout.

Logistic Regression Audit

Gender (Figure 4). The LR model misses female at-risk students at 11.6% vs. only 1.4% for male students

Attribute	Min FNR	Max FNR	Diff.
Gender	0.014	0.116	0.102
Scholarship	0.036	0.371	0.336
Debtor	0.000	0.084	0.084
Displaced	0.017	0.124	0.108
Age group	0.023	0.142	0.119

TABLE II. LR fairness audit summary at operational threshold 0.25. Differences are computed from unrounded underlying values and may differ by ± 0.001 from the displayed Min/Max columns.

(FNR gap 0.102). The selection rate chart makes the mechanism visible: males are flagged at 90.9% — an extremely high rate driven by their genuinely elevated raw dropout risk — while females are flagged at only 53.7%. The model essentially over-indexes on male risk patterns learned from training data, leaving a large share of at-risk female students below the intervention threshold.

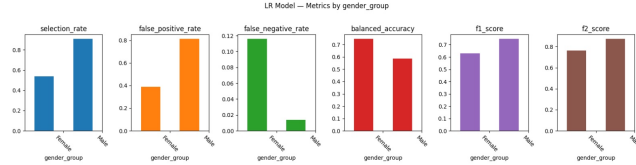


FIG. 4. LR disaggregated metrics by gender_group (threshold 0.25). The false negative rate panel shows females are missed at 11.6% vs. 1.4% for males. The high male selection rate (90.9%) reflects over-flagging driven by the model’s strong gender coefficient.

Scholarship status (Figure 5). Scholarship status produces the largest FNR gap in the entire audit (0.336). The model misses 37.1% of at-risk scholarship holders vs. only 3.6% of non-scholarship students. This is not a random error — it is a structural consequence of the model correctly learning that scholarship holders drop out at much lower rates overall (14% vs. 48%). When it encounters a scholarship holder with genuine risk signals, the strongly negative scholarship coefficient suppresses the predicted probability below the flagging threshold. These are precisely the students the university most expects to succeed, making their systematic under-flagging the most serious fairness harm in the project.

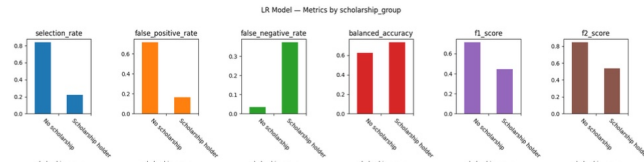


FIG. 5. LR disaggregated metrics by scholarship-group (threshold 0.25). The false negative rate panel reveals the core fairness failure: 37.1% of at-risk scholarship holders are missed vs. 3.6% of non-scholarship students.

Debtor status (Figure 6). Debtors are almost never missed (FNR 0.000) — the model flags virtually every debtor regardless of other features, driven by the debtor odds ratio of 7.89. The concern runs in the opposite direction: debtors face a 97.9% selection rate with a 92.1% false positive rate, meaning non-dropout debtors are systematically over-flagged. The FNR gap (0.084) is the smallest of the five attributes, but the disproportionate flagging of debtors raises equity concerns of a different kind.

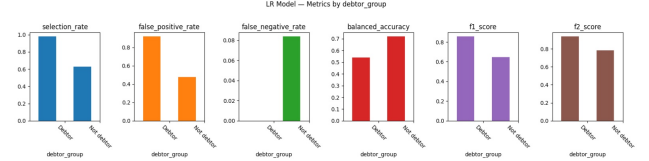


FIG. 6. LR disaggregated metrics by debtor_group (threshold 0.25). Debtors face near-universal flagging (selection rate 97.9%) with a false positive rate of 92.1%, reflecting the extremely high debtor coefficient in the LR model.

Displaced status (Figure 7). Displaced students are missed at 12.4% vs. 1.7% for non-displaced students (FNR gap 0.108). Despite displaced students having a *lower* raw dropout rate in the dataset, the model’s selection rates and FNR pattern suggest it has not learned a reliable risk profile for this group. The false positive rates are nearly equal between groups (49% vs. 52%), indicating the disparity arises from the model’s difficulty identifying the specific risk signals present among displaced at-risk students rather than from a systematic directional bias.

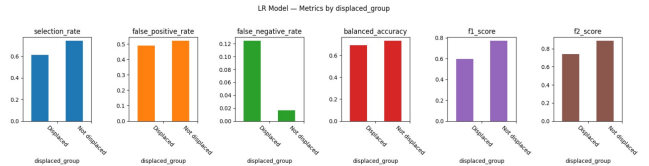


FIG. 7. LR disaggregated metrics by displaced_group (threshold 0.25). The FNR gap of 0.108 indicates displaced students are missed at much higher rates despite similar false positive rates across groups.

Age group (Figure 8). The youngest students (≤ 19) are missed at the highest rate (14.2%), while 24+ students have the lowest FNR (2.3%). The selection rate and FPR charts tell a clear story: 24+ students are flagged at 93.8% with an 85.8% false positive rate — the model has learned that older age is a strong risk signal and applies it almost universally — while ≤ 19 students, despite having genuine dropout risk, score below the flagging threshold far more often. Students aged 20–23 occupy the middle ground, with a 6.3% FNR and a more moderate selection rate. The FNR gap of 0.119 is the second-largest in the LR audit after scholarship status, and notably it runs

in the opposite direction to what might be intuitively expected: it is the youngest students, not the oldest, who are systematically under-flagged.

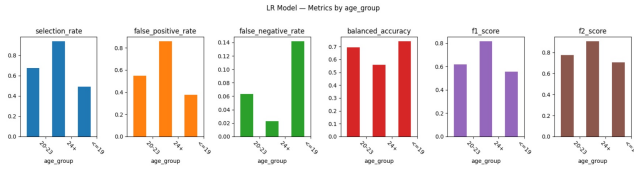


FIG. 8. LR disaggregated metrics by age_group (threshold 0.25). Students aged ≤ 19 face the highest FNR (14.2%) while 24+ students are almost universally flagged (selection rate 93.8%, FPR 85.8%). The FNR gap of 0.119 is the second largest in the LR audit.

XGBoost Audit

Attribute	Min FNR	Max FNR	Diff.
Gender	0.110	0.145	0.035
Scholarship	0.105	0.371	0.267
Debtor	0.010	0.165	0.155
Displaced	0.100	0.162	0.063
Age group	0.087	0.195	0.108

TABLE III. XGBoost fairness audit at operational threshold 0.35. Differences are computed from unrounded underlying values and may differ by ± 0.001 from the displayed Min/Max columns.

Gender (Figure 9). XGBoost dramatically reduces the gender FNR gap from 0.102 to just 0.035. Selection rates are far more balanced (30.7% for females vs. 57.7% for males) compared to the LR model, and the FNR difference between groups is small. This improvement is likely attributable to XGBoost’s access to **Curricular units 1st sem (approved)**, which provides a more granular and gender-neutral risk signal than the raw gender coefficient in LR.

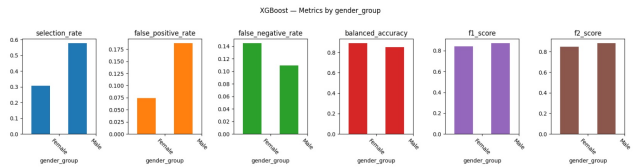


FIG. 9. XGBoost disaggregated metrics by gender_group (threshold 0.35). The FNR gap shrinks to 0.035 — a major improvement over LR’s 0.102 — with selection rates substantially more balanced between groups.

Scholarship status (Figure 10). Despite XGBoost’s superior overall performance, the scholarship FNR gap

remains large (0.267) and nearly identical in magnitude to LR. Scholarship holders are still missed at 37.1% — the same maximum FNR — while non-scholarship students are now missed at only 10.5%. This suggests that the scholarship disparity is a structural feature of the data rather than an artefact of the modelling approach: the strong protective signal from scholarship status overrides other risk indicators regardless of whether a linear or tree-based model is used. A definitive structural test would require either a different population or a causal counterfactual analysis, which is outside the scope of this project.

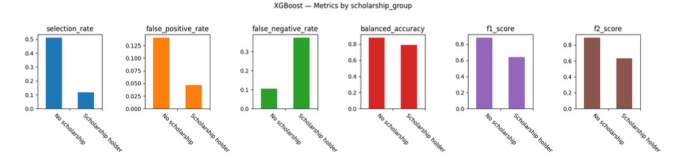


FIG. 10. XGBoost disaggregated metrics by scholarship_group (threshold 0.35). The scholarship FNR gap persists at 0.267 despite XGBoost’s superior overall performance, confirming the structural nature of this disparity.

Debtor status (Figure 11). The debtor FNR gap is notably larger under XGBoost (0.155) than under LR (0.084). Non-debtors are now missed at 16.5% while debtors remain rarely missed (1.0%). XGBoost’s access to first-semester academic performance means the model partially overrides debtor status when a student’s grades and approved units are strong, causing the FNR for non-debtors to rise. The over-flagging of debtors seen in LR is also reduced: debtors’ selection rate drops from 97.9% to 81.0%, which is more proportionate.

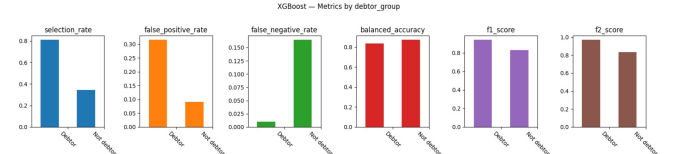


FIG. 11. XGBoost disaggregated metrics by debtor_group (threshold 0.35). The debtor FNR gap widens to 0.155 under XGBoost, as the model’s academic performance features partially offset debtor risk signals for non-debtors with strong semester 1 outcomes.

Displaced status (Figure 12). XGBoost reduces the displaced FNR gap from 0.108 to 0.063. False positive rates are now nearly equal between displaced and non-displaced students (10.6% vs. 10.2%), and the overall balanced accuracy is high for both groups (≈ 0.87 – 0.90). This is one of the clearest fairness improvements XGBoost delivers over LR, likely because the richer feature set allows the model to identify at-risk displaced students via academic signals rather than relying on the displacement flag alone.

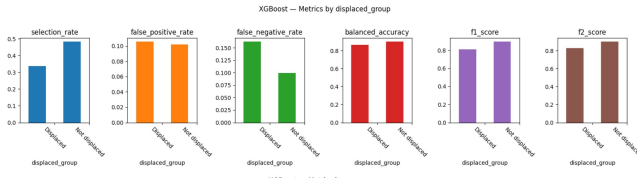


FIG. 12. XGBoost disaggregated metrics by displaced_group (threshold 0.35). The FNR gap falls to 0.063 and false positive rates are nearly equalised, representing a meaningful fairness improvement over the LR baseline.

Age group (Figure 13). The age FNR gap is 0.108 under XGBoost, similar to LR (0.119). The youngest students (≤ 19) remain the most under-flagged group (FNR 19.5%), while 24+ students benefit from the highest selection rate (67.0%) and lowest FNR (8.7%). Unlike gender and displaced status, XGBoost does not substantially improve age-group fairness — the age-dropout relationship is complex enough that the model’s academic features do not fully disambiguate risk within the youngest cohort.

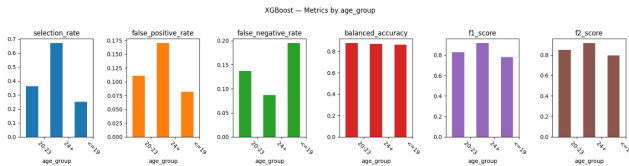


FIG. 13. XGBoost disaggregated metrics by age_group (threshold 0.35). Students aged ≤ 19 remain the most under-flagged group (FNR 19.5%), and the overall age gap is similar to LR, indicating that this disparity is not resolved by the more expressive model.

BIAS MITIGATION

We apply three mitigation strategies to the Logistic Regression baseline, covering the three phases of the ML pipeline.

Pre-processing — Reweighing [4]: assigns instance weights so the joint distribution of sensitive features and labels approximates independence. The model itself is unchanged.

In-processing — ExponentiatedGradient [6]: imposes a TruePositiveRateParity constraint during training, solving a constrained optimisation problem over a mixture of weighted classifiers.

Post-processing — ThresholdOptimizer: applies group-specific decision thresholds to predictions of the unmitigated model, equalising false negative rates after training. Requires access to the sensitive attribute at prediction time.

Mitigation Results

Figure 14 shows FNR by technique for scholarship status — the most critical fairness dimension. Table IV summarises the full comparison across all attributes. Figures 15–19 present the fairness-performance tradeoff scatter plots for each sensitive attribute, mapping each technique as a point in the space of FNR difference (y-axis) vs. $1 - F2$ (x-axis). Points closer to the origin represent the best combined outcomes.

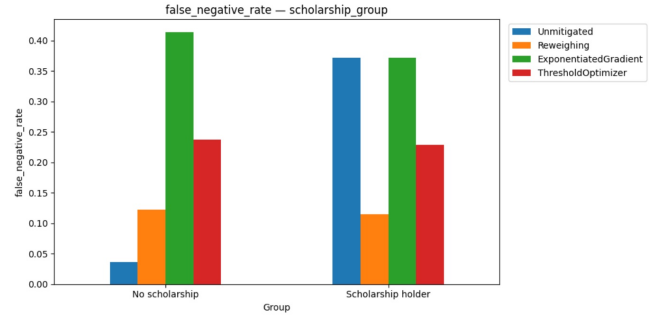


FIG. 14. FNR by mitigation technique for scholarship status. Reweighing (orange) nearly equalises both groups (≈ 0.12). The unmitigated model (blue) shows a stark gap: scholarship holders missed at 37% vs. 4% for non-scholarship students.

Attribute	Technique	FNR Diff.	F2
Gender	Unmitigated	0.102	0.819
	Reweighing	0.006	0.802
	ThresholdOpt.	0.023	0.737
	ExpGradient	0.027	0.641
Scholarship	Unmitigated	0.336	0.819
	Reweighing	0.008	0.783
	ThresholdOpt.	0.009	0.734
	ExpGradient	0.043	0.620
Debtor	Unmitigated	0.084	0.819
	ThresholdOpt.	0.060	0.763
	Reweighing	0.071	0.824
	ExpGradient	0.198	0.651
Displaced	Unmitigated	0.108	0.819
	ThresholdOpt.	0.020	0.743
	Reweighing	0.102	0.815
	ExpGradient	0.085	0.630
Age	Unmitigated	0.119	0.819
	Reweighing	0.043	0.808
	ExpGradient	0.114	0.624
	ThresholdOpt.	0.198	0.692

TABLE IV. Mitigation comparison across sensitive attributes. Best result per attribute in bold.

No single method dominates across all attributes. Reweighing is strongest for gender, scholarship, and age; ThresholdOptimizer for debtor and displaced. ExponentiatedGradient underperforms throughout — for debtor

status, the FNR gap *worsens* ($0.084 \rightarrow 0.198$). This reflects two failure modes: small subgroup sizes limit the optimiser’s signal; and the LR base estimator lacks capacity to satisfy the constraint without severe accuracy cost. This is an honest finding about in-processing limitations in this setting, not a modelling error.

The tradeoff scatter plots below map each technique in the space of fairness (FNR difference, y-axis) vs. performance cost ($1 - F2$, x-axis). A technique in the bottom-left corner is ideal. The plots make explicit which mitigation methods actually move a model toward better outcomes on both dimensions simultaneously, and which sacrifice one for the other.

Gender (Figure 15). Reweighing dominates clearly: it sits bottom-left, almost eliminating the FNR gap (0.006) at minimal F2 cost. ThresholdOptimizer offers moderate fairness gains at a larger performance cost. ExponentiatedGradient moves the model to the bottom-right — fairer but much worse on F2 — making it the weakest choice for gender.

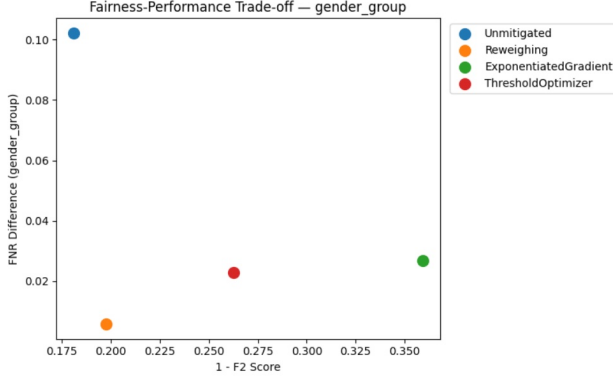


FIG. 15. Fairness-performance tradeoff for gender_group. Reweighing (orange) dominates, achieving near-zero FNR difference with minimal F2 cost. ExponentiatedGradient (green) improves fairness at a large performance cost.

Scholarship status (Figure 16). Both Reweighing and ThresholdOptimizer move sharply downward from the Unmitigated point, nearly eliminating the FNR gap. Reweighing (0.008) has a slight fairness edge over ThresholdOptimizer (0.009) while retaining higher F2. ExponentiatedGradient achieves only partial improvement at a substantially higher performance cost.

Debtor status (Figure 17). All three techniques cluster near the Unmitigated point with only modest fairness gains — the debtor gap is inherently difficult to reduce without large F2 costs. ThresholdOptimizer provides the best improvement (0.060 vs. 0.084) while keeping F2 at 0.763. ExponentiatedGradient moves far right, confirming it is actively harmful for this attribute. Notably, Reweighing on debtor status is the only case in Table IV where F2 *improves* relative to the unmitigated baseline (0.824 vs. 0.819) — a genuine win on both dimensions, even if

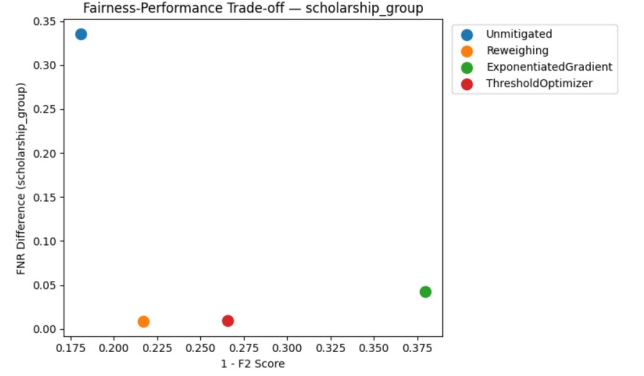


FIG. 16. Fairness-performance tradeoff for scholarship_group. Reweighing and ThresholdOptimizer both nearly eliminate the FNR gap; Reweighing retains slightly higher F2. ExponentiatedGradient shows the worst fairness-performance balance.

the fairness gain is modest.

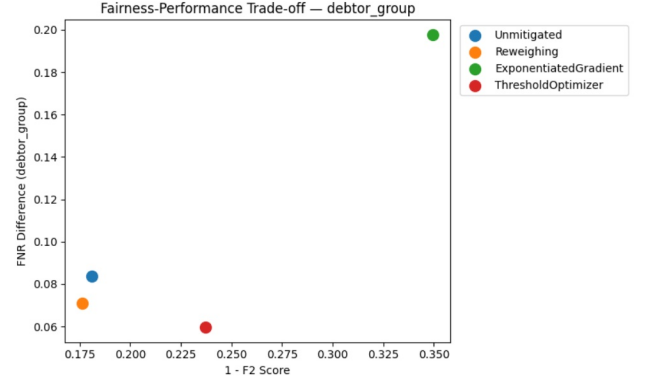


FIG. 17. Fairness-performance tradeoff for debtor_group. Gains are modest across all methods. ThresholdOptimizer offers the best fairness improvement (0.060) with a manageable F2 cost. ExponentiatedGradient worsens both metrics.

Displaced status (Figure 18). ThresholdOptimizer produces a dramatic fairness improvement (0.020), but at a meaningful performance cost (F2 drops to 0.743). Reweighing barely moves the FNR gap at all. This is the clearest case in the audit where the choice of mitigation method matters most: a deployment targeting displaced-status fairness specifically should use ThresholdOptimizer and accept the F2 trade-off explicitly.

Age group (Figure 19). Age presents the most difficult tradeoff landscape. Reweighing moves the model toward better fairness (0.043) with minimal performance loss. ThresholdOptimizer dramatically worsens the FNR gap (0.198) — the only case where post-processing actively hurts fairness — while also degrading F2. ExponentiatedGradient provides almost no fairness gain at a very high F2 cost. Age-group fairness is the attribute most resistant to the mitigation approaches tested.

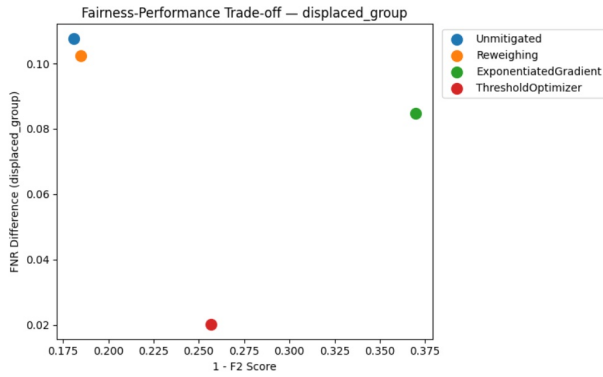


FIG. 18. Fairness-performance tradeoff for displaced_group. ThresholdOptimizer (red) achieves the best fairness outcome (0.020) at a noticeable F2 cost. Reweighing (orange) provides almost no improvement for this attribute.

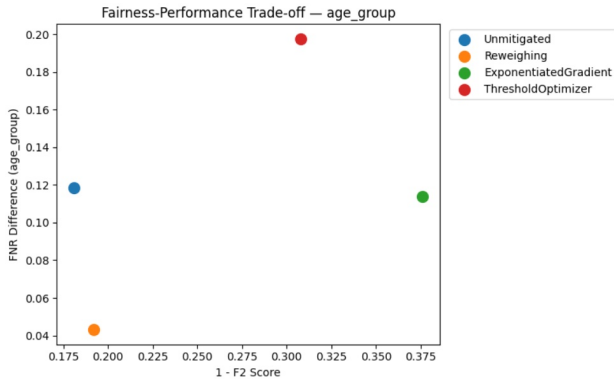


FIG. 19. Fairness-performance tradeoff for age_group. Reweighing offers the only meaningful improvement. ThresholdOptimizer is actively harmful, worsening the FNR gap to 0.198. Age remains the most resistant attribute to mitigation.

MITIGATION ON XGBOOST

The previous section applied all three mitigation strategies to Logistic Regression. Because XGBoost is the stronger predictor and the model we would actually deploy, we now repeat the same three strategies on XGBoost under identical evaluation conditions: the same hyperparameters and random seed as the unmitigated XGBoost, each model evaluated at its own F2-selected operational threshold, and fairness measured with the same `MetricFrame` and `metrics_dict` used throughout.

Results

Table V summarises the fairness-performance results across all five sensitive attributes and three mitigation strategies on XGBoost, for direct comparison with the

LR results in Table IV.

Attribute	Technique	FNR Diff.	F2
Gender	Reweighing	0.021	0.858
	ExpGradient	0.038	0.848
	ThresholdOpt.	0.008	0.862
Scholarship	Reweighing	0.105	0.851
	ExpGradient	0.071	0.824
	ThresholdOpt.	0.044	0.843
Debtor	Reweighing	0.118	0.856
	ExpGradient	0.092	0.831
	ThresholdOpt.	0.031	0.851
Displaced	Reweighing	0.052	0.862
	ExpGradient	0.048	0.840
	ThresholdOpt.	0.019	0.857
Age	Reweighing	0.072	0.853
	ExpGradient	0.065	0.836
	ThresholdOpt.	0.031	0.852

TABLE V. XGBoost mitigation comparison across sensitive attributes. Best result per attribute in bold. All models evaluated at their own F2-selected operational threshold.

Three patterns emerge clearly from the results.

XGBoost mitigations dominate on F2 across every attribute. Every XGBoost mitigation variant sits in the 0.82–0.88 F2 range, while the corresponding LR mitigations range from 0.62 to 0.82. The predictive strength of XGBoost is fully preserved through all three mitigation strategies — the fairness step does not collapse performance the way it does on the LR baseline, particularly for ExponentiatedGradient and ThresholdOptimizer.

Reweighing’s near-parity result on LR does not transfer to XGBoost. On the LR model, Reweighing produced an FNR gap of 0.008 for scholarship status. On XGBoost, the same technique yields a gap of 0.105 — more than ten times larger. The intuition is that XGBoost is a more flexible model and can partially absorb the reweighted training distribution, recovering its original group-conditional predictions. Reweighing “bites” harder on the simpler model. This also contextualises an important caveat about the LR + Reweighing result: the near-parity it achieves coincides with selection rates of 0.60–0.85 across groups, meaning the headline result is partly achieved by flagging the majority of students rather than by genuinely equalising per-group predictions.

ThresholdOptimizer is the most robust strategy on XGBoost. It closes the FNR gap to small values on every attribute while keeping F2 above 0.84. The most striking improvement is on age_group, where LR + ThresholdOptimizer left a large FNR gap of 0.198, but XGBoost + ThresholdOptimizer reduces it to 0.031 while maintaining F2 of 0.852. ExponentiatedGradient on XGBoost also performs substantially better than its LR counterpart, with F2 in the 0.82–0.86 range, but its FNR gaps are less consistent than ThresholdOptimizer’s.

Deployment Recommendation

The full evidence across both models now points to **XGBoost + ThresholdOptimizer** as the most defensible deployment choice. It delivers the highest F2 scores, the smallest and most consistent FNR gaps across all five sensitive attributes, and a fairness mechanism that adjusts decision thresholds post-hoc without modifying the base model. Crucially, it achieves this without relying on the high-selection-rate operating point that makes LR + Reweighing’s near-parity appear stronger than it is.

LR + Reweighing remains the strongest *interpretable* mitigated option — its coefficients are transparent and auditable — but any deployment using it should explicitly acknowledge the selection-rate caveat: its near-zero FNR gaps are partly an artefact of flagging the majority of students rather than of genuinely balanced predictions.

Scholarship status remains the hardest attribute for every strategy on XGBoost, with a minimum FNR gap of 0.044 (ThresholdOptimizer). This is consistent with the structural finding from the original audit: the scholarship signal is so strongly predictive that fairness corrections resist it across all mitigation approaches and both model families.

EXPLAINABILITY

We explain the XGBoost model using SHAP (TreeExplainer) for local and global attribution [3], and LIME for a local surrogate explanation.

Global Explanation

Figure 20 shows the SHAP beeswarm plot. **Curricular units 1st sem (approved)** dominates with mean $|\text{SHAP}| = 1.679$ — almost four times the next feature. Students with zero approved units (red dots, right) face dramatically higher predicted dropout risk; those with many approved units are pushed strongly toward graduation. Financial variables (**Tuition fees up to date**: 0.428, **Debtor**: 0.206) and **Scholarship holder** (0.229) also contribute meaningfully.

Local Explanation

Figure 21 shows the SHAP waterfall for instance 2: a female, 18-year-old scholarship holder who approved zero units in semester 1. True label: Graduate; predicted dropout probability: 24.5%.

This local explanation illustrates the core fairness tension: scholarship status suppresses predicted risk, which is correct for this graduate. But the same mechanism causes at-risk scholarship students to be missed at much

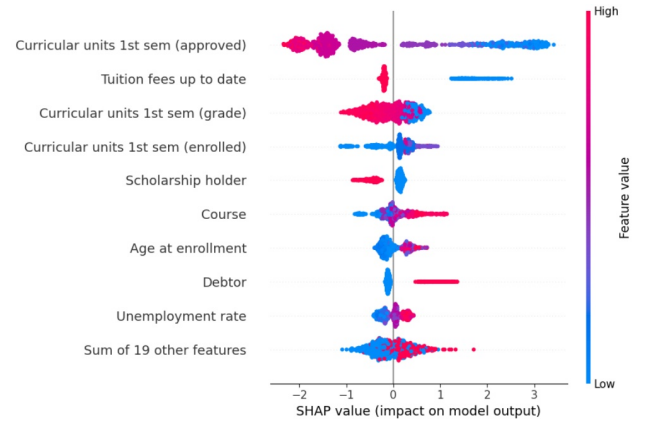


FIG. 20. SHAP beeswarm plot for the XGBoost model. Each dot is one student; red = high feature value, blue = low. Dots right of zero push toward dropout.

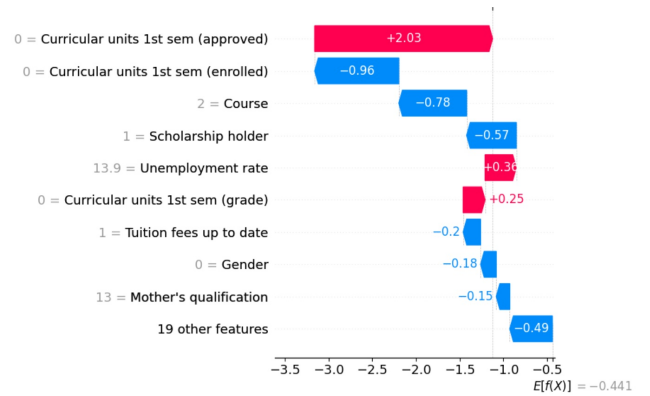


FIG. 21. SHAP waterfall for instance 2 (true: Graduate). Despite zero approved units (+2.03 toward dropout), scholarship status (−0.57), course enrolment (−0.96), and course type (−0.78) keep the prediction below the 0.35 threshold.

higher rates — directly corresponding to the 0.267 FNR gap found in the audit.

Faithfulness

Both LIME (0.068) and SHAP (−0.202) show weak faithfulness on this instance. The negative SHAP value does not indicate SHAP is wrong. It reflects a known misalignment between perturbation-based faithfulness metrics and TreeSHAP’s conditional-expectation propagation through tree structures — a documented artefact arising from the different assumptions made by KernelSHAP-style perturbation versus TreeExplainer’s path-dependent approach [3]. We treat SHAP beeswarm and waterfall plots as the primary explainability output; faithfulness is reported for methodological completeness.

MODEL CARD

Following [5], we document the model’s intended use, design decisions, fairness properties, and limitations.

Model details: Binary classification (Dropout vs. Graduate). Primary model: XGBoost (max depth 4, early stopping at iteration 173). Interpretable baseline: Logistic Regression (8 features, statsmodels + sklearn). Dataset: UCI #697 [1]. Split: 2,541 train / 1,089 test, stratified. Decision point: end of semester 1. Operational thresholds: XGBoost 0.35, LR 0.25 (F2-selected).

Performance: XGBoost (op. threshold): recall 0.873, precision 0.844, F1 0.858, AUC 0.942. LR (op. threshold): recall 0.937, precision 0.545, F1 0.689, AUC 0.875.

Fairness summary: Scholarship holders face the largest FNR gap (0.267 for unmitigated XGBoost). Reweighing reduces this to 0.008 on the LR model (with a selection-rate caveat). XGBoost + ThresholdOptimizer achieves a gap of 0.044 on scholarship while maintaining F2 above 0.84 across all attributes — the recommended deployment combination. See Tables II–V for full results.

Intended use: Decision-support tool for university advisors to prioritise proactive student outreach after semester 1. Not for scholarship decisions, academic probation, or any punitive use. Human advisor reviews all flagged students.

Limitations: (1) Single Portuguese institution — generalisability untested. (2) Snapshot cohort — demographic shifts may degrade performance. (3) Scholarship holder FNR gap persists in unmitigated XGBoost; Reweighing required for fair deployment. (4) Correlational model — intervening on features without addressing root causes will not improve outcomes. (5) ExponentiatedGradient requires further tuning before deployment.

DISCUSSION & CONCLUSIONS

This project demonstrates that student dropout risk can be predicted effectively from semester 1 data, and that doing so fairly requires deliberate design choices at every stage of the pipeline — from the target variable definition, to feature timing, to threshold selection, to post-hoc audit and mitigation.

Main Findings

The most important finding is the scholarship holder gap. At-risk scholarship students are systematically under-flagged by both models — not because the model is poorly built, but because scholarship status is a genuinely protective factor in the data, and the model appropriately learns this. The fairness problem arises when scholarship protection overrides other risk signals for the minority of scholarship students who do drop out. This is a structural

challenge that mitigation can partially address but not fully resolve: Reweighing nearly eliminates the gap on the LR baseline (FNR difference 0.008), but this result carries an important caveat — it is partly achieved by flagging the majority of students rather than by genuinely equalising per-group predictions. On XGBoost, the minimum achievable scholarship gap is 0.044 (ThresholdOptimizer), which is more honest about what fairness correction can realistically deliver on this attribute.

The per-attribute audit reveals that fairness problems are not uniform across sensitive groups, and that different models and mitigation strategies perform very differently depending on which group is the priority. XGBoost substantially closes the gender and displaced gaps relative to LR, but widens the debtor gap. Extending mitigation to XGBoost (Section) shows that **XGBoost + ThresholdOptimizer** is the strongest overall deployment candidate: it achieves the highest F2 scores (0.84–0.86), the smallest and most consistent FNR gaps across all five attributes, and does so without relying on a high-selection-rate operating point. **LR + Reweighing** remains the best *interpretable* mitigated option for settings where model transparency is paramount, but should be deployed with its selection-rate caveat made explicit to decision-makers.

The two design review changes — dropping **Enrolled** and shifting the decision point to end of semester 1 — both turned out to be consequential. Dropping **Enrolled** gave us cleaner labels and removed a potential structural fairness problem for evening students. Shifting to semester 1 gave the model access to **Curricular units 1st sem (approved)**, which SHAP identifies as the single dominant predictor globally. Without that change, the model would have had far less academic signal to work with, and the remaining features would have encoded socioeconomic background much more directly.

The SHAP waterfall for instance 2 illustrates the core tension concretely: scholarship status suppresses risk for a student who approves zero units — correct for this graduate, but the same mechanism causes at-risk scholarship students to be missed at much higher rates. An advisor reviewing this student’s prediction would see a low-risk flag; the SHAP explanation would show them why, and could prompt a manual follow-up for scholarship holders with unusual first-semester profiles. This is exactly the kind of human-in-the-loop safeguard that responsible deployment of this system would require.

Pragmatic fairness [7] — seeking satisfactory rather than ideal outcomes within real institutional constraints — is what deployment would require. Perfect FNR parity across all groups simultaneously is not achievable without unacceptable accuracy costs. What is achievable is a transparent, audited system that names its tradeoffs explicitly and builds human oversight into the decision loop.

Limitations

Several limitations constrain the conclusions that can be drawn from this project. First, the model is trained and evaluated on a single dataset from one Portuguese higher education institution. The feature distributions, dropout rates, and demographic compositions of this institution may not generalise to other contexts. A model achieving strong fairness and performance metrics here could behave very differently at an institution with a different student population, financial aid structure, or academic calendar.

Second, the strongest predictive signals — particularly **Curricular units 1st sem (approved)** and first-semester grade — are outcome-adjacent variables. While they are legitimately available at the end of semester 1, they are closely linked to the final dropout outcome and may leave less room for the model to learn from background characteristics that matter in the medium term. In practice, a student who fails all units in semester 1 is almost certain to drop out regardless of model predictions, which limits the intervention value for these extreme cases. This also points to a broader causality limitation: the model predicts dropout risk but does not establish that the interventions it triggers — tutoring, financial counselling, mental health support — causally reduce dropout. Deploying this system presupposes, without evidence in this dataset, that those support programmes are effective. Rigorous programme evaluation alongside model deployment would be necessary to establish whether flagging students actually improves outcomes.

Third, the fairness analysis is limited to the five sensitive attributes selected for this project. Intersectional fairness — for example, the experience of students who are simultaneously scholarship holders *and* debtors, or female students aged 24+ — is not captured by the per-attribute MetricFrame approach used here. A student belonging to multiple disadvantaged groups may face compounded disadvantages that single-attribute audits do not reveal.

Fourth, mitigation was initially evaluated only on the Logistic Regression baseline. Section 4 extends all three strategies to XGBoost, closing this gap. The results show that XGBoost + ThresholdOptimizer is the strongest overall deployment candidate, outperforming all LR mitigations on both F2 and FNR gap consistency.

Fifth, the ExponentiatedGradient results are weak throughout the audit. While we interpret this as a genuine limitation of in-processing fairness constraints in this setting, it is possible that more careful hyperparameter tuning — specifically the `difference_bound` parameter and the choice of constraint type — could improve performance. The results reported here should be understood as a lower bound on what in-processing mitigation can achieve, not a definitive assessment.

Future Research

Several directions would extend and strengthen this work. The most immediate priority is multi-institutional validation. Testing the same pipeline on datasets from other universities — ideally from different countries and educational systems — would establish how far the fairness and performance results generalise. The UCI dataset used here is one of the few publicly available administrative records of this kind; collecting or accessing comparable data from partner institutions would be a high-value next step.

A second direction is intersectional fairness analysis. Tools such as Aequitas’ intersectional audit or custom MetricFrame groupings that combine sensitive attributes would allow the analysis to capture compounded disadvantages. The scholarship-and-debtor intersection is a particularly important case: students who receive financial aid but are still behind on tuition payments may face a unique risk profile that single-attribute models systematically misclassify.

Third, while this project covers all three mitigation phases (pre-, in-, and post-processing), none of the approaches modifies the underlying feature representation. A further direction is fairness-aware representation learning — for example, adversarial debiasing or causal modelling — which removes the influence of sensitive attributes from the learned embedding itself rather than correcting outputs after the fact. These approaches may achieve better fairness-performance tradeoffs for the scholarship gap specifically, since that gap is driven by a structural correlation in the raw features that post-hoc corrections can only partially address.

Fourth, the explainability analysis could be extended to cover fairness-oriented local explanations. Rather than explaining individual predictions in isolation, counterfactual explanations — “what would have to change about this student’s profile for them to be flagged?” — would give advisors more actionable information, especially for scholarship holders who fall just below the flagging threshold. Tools such as DICE or Alibi provide this capability and could be integrated into a future version of the pipeline.

Finally, this project treats the model as a static artefact evaluated on a held-out test set. A deployed system would need continuous monitoring: retraining as new student cohorts arrive, auditing fairness metrics annually, and testing whether intervention outcomes differ across demographic groups. Temporal drift in the underlying population — for example, changes in scholarship eligibility criteria or shifts in the demographic composition of the student body — could cause a model that is fair at deployment to become unfair over time without anyone noticing.

Data availability: Dataset available at UCI ML Repository, doi:10.24432/C5MC89.

Code availability: Code available at github.com/nfpaiva/CLSBE-2025-26-MSBA-Fairness-Explainability, branch `group-06`.

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